Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	25 September 2026
Team ID	
Project Name	Medical Diagnosis Center by Service Portal (ServiceNow)
Maximum Marks	4 Marks

Technical Architecture:

This document describes the proposed technology stack and technical architecture for the Medical Diagnosis Center application developed using the ServiceNow Service Portal.

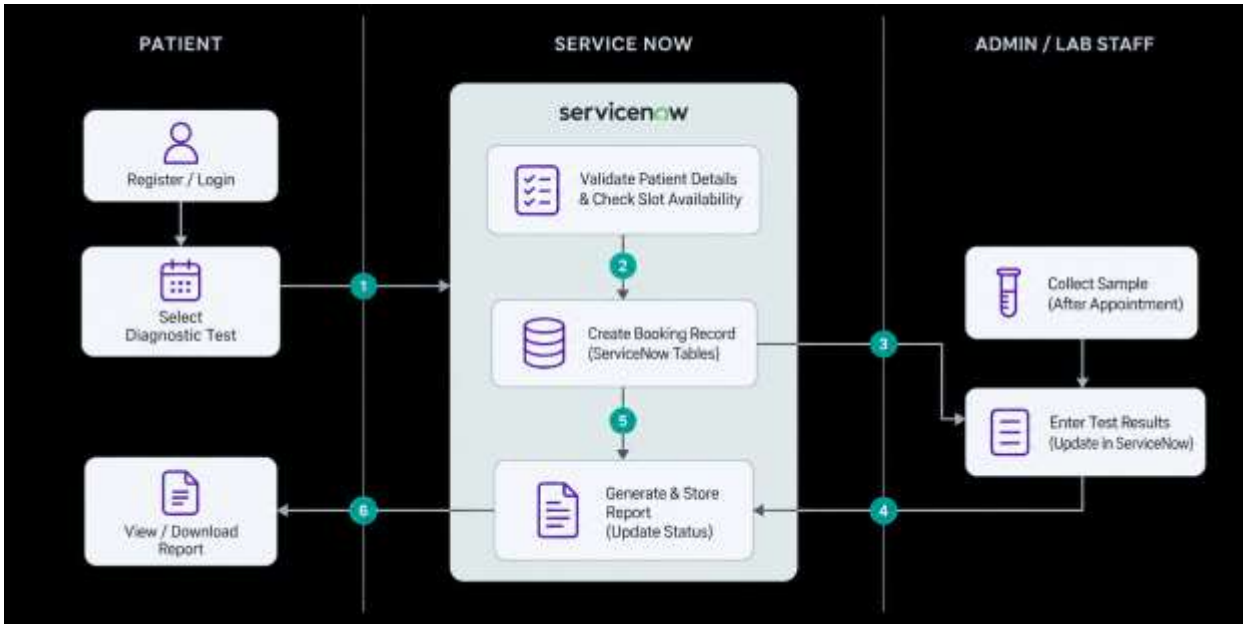


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Portal pages for patients, lab staff, reception staff, and administrators. Includes test catalog, booking, appointments, and report access.	ServiceNow Service Portal, HTML Application, CSS Codes, JavaScript / JSAPI
2	Application Logic – Patient and Booking	Register patients, validate form inputs, create bookings, check appointment details, and prevent conflicting or overlapping bookings.	ServiceNow server-side JavaScript, Business Rules, Script Includes
3	Application Logic – Workflow	Manage booking and diagnostic status, including requested, pending approval, confirmed, sample collected, processing, completed, and report available.	ServiceNow Flow Designer, workflow configuration, Business Rules

4	Application Logic – Administration	Allow authorized staff to manage diagnostic tests, patient records, bookings, results, and access permissions.	ServiceNow Admin Tools, tables, forms, roles, and ACLs
5	Database	Store patient information, diagnostic test details, appointment records, results, and report references.	ServiceNow platform tables and fields
6	Cloud Database	ServiceNow-hosted platform data storage for application records.	ServiceNow-managed cloud platform
7	File Storage	Store diagnostic reports and related files for authorized viewing or download.	ServiceNow attachments / platform file storage

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	The supplied skills list names HTML, CSS, JSAPI, XML concepts, and JSON. No additional open-source framework is specified.	HTML, CSS, JavaScript / JSAPI, XML, JSON
2	Security Implementations	Restrict patient and report information to authorized roles. Apply access controls and validate user input on the server side.	ServiceNow roles, ACLs, authentication, server-side validation
3	Scalable Architecture	Portal interface, application logic, and platform data are separated into presentation, application, and data layers.	ServiceNow Service Portal and platform services
4	Availability	Application availability depends on the ServiceNow instance and its service availability. No separate load balancer or multi-server setup is specified.	ServiceNow-hosted platform
5	Performance	Use efficient table queries, server-side validation, and suitable form/widget design. Test portal pages, booking operations, and report access.	ServiceNow tables, Business Rules, Service Portal widgets